

A Conditional Native-Qutrit Latch with an Explicit Damped Auxiliary Qubit

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Abstract—We replace an ideal terminal jump by an explicit interaction between a native qutrit and a damped auxiliary qubit. The retained six-state model exposes finite reservoir response, reverse jumps and interaction-norm overhead. Directed interval arithmetic certifies a cold retained-channel error below 2^{-16} over a declared readout window for three frozen parameter choices. A conservative perturbation argument extends one case to a nonzero reverse-rate interval. Exact stationary identities distinguish clock-only capture from auxiliary reset, while encoded Pauli coefficients show that native subsystem locality does not establish qubit two-locality. Completion ingress, microscopic bath validity and physical implementation remain open.

Index Terms—Autonomous capture, engineered dissipation, local interactions, reverse jumps, MQM.

I. ADMISSION CONTRACT AND RETAINED DOMAIN

The preceding capacity-boundary latch uses $g = 1$ and ideal jump $\sqrt{\Gamma}|d\rangle\langle e|$, with $\Gamma = 2\sqrt{2}$. Its error certificate assumes trusted saturation and completion eligibility, product terminal launch $|m\rangle$, identity coupling to stored data and a traced Markov bath. This successor specifies the reservoir interface rather than presuming that the ideal jump is a physical circuit element.

Let C be a native qutrit with levels m, e, d , and R a physical auxiliary qubit with levels $0, 1$. The retained latch domain has dimension $3 \cdot 2 = 6$. Stored data B and arbitrary reference spectators are retained and all latch terms act as I_B . Launch is $|m, 0\rangle$. The target is $|d, 0\rangle$, including auxiliary reset. No stored-data Hamiltonian is included. The continuum bath is still traced, so the model does not claim a finite closed-system absorbing state.

Eligibility is an input contract. In the cell, coherent m - e entry is enabled only for previously admitted episodes; constructing this switch autonomously from completion and occupancy registers remains OPEN. Closing that gate cannot be inferred from the local reservoir pair. No source-defined occupancy fraction is identified with transmission, bath temperature or a jump rate.

II. MICROSCOPIC-CELL SPECIFICATION AND LOCALITY

In an interaction picture, define $S = |d\rangle\langle e|$ and

$$H_{CR} = g(|m\rangle\langle e| + \text{h.c.}) \otimes I_R + \lambda(S \otimes \sigma_R^+ + S^\dagger \otimes \sigma_R^-), \quad (1)$$

$$J_\downarrow = \sqrt{\kappa}I_C \otimes \sigma_R^-, \quad J_\uparrow = \sqrt{\kappa r}I_C \otimes \sigma_R^+. \quad (2)$$

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The six-state master equation is $\dot{\rho} = -i[H_{CR}, \rho] + \mathcal{D}[J_\downarrow]\rho + \mathcal{D}[J_\uparrow]\rho$. Here r is a phenomenological reverse/downward rate ratio. It is not a calibrated material temperature. This is an explicit damped two-level auxiliary, not an exact harmonic-reservoir pseudomode construction. Its finite response produces memory in the reduced qutrit dynamics.

Adiabatic elimination is an established approximate method for open systems [1]. In the fast cold-auxiliary limit, eliminating $|d, 1\rangle$ gives $\Gamma_{\text{eff}} \simeq 4\lambda^2/\kappa$. Our frozen matching rule is $\lambda = \sqrt{\Gamma\kappa}/2$, with $\kappa \in \{2, 8, 32, 128\}$ and $\Gamma = 2\sqrt{2}$. Matching this scalar is disclosed engineering; it does not certify the elimination approximation at every value. The finite model is evaluated directly.

The interaction is two-local across a native qutrit–qubit pair. It is not a qubit two-local compiler. Encode $m = 00, e = 01, d = 10$ in two physical qubits and use a third for R . The exchange becomes

$$\frac{\lambda}{4}(XXX + YYX - YXY + XYY). \quad (3)$$

Its XXX coefficient is $\lambda/4 \neq 0$, so unrestricted equality to a sum of one- and two-qubit Pauli terms fails in this encoding. This is an obstruction to the displayed encoding, not to all possible dilations or gadget constructions. The additional unused state 11 must also be included in a qubit compiler's leakage ledger.

III. FROZEN TESTS AND COLD READOUT CERTIFICATE

Before evaluation, freeze $r \in \{0, 2^{-20}, 2^{-16}, 2^{-12}\}$ and times $4, 16, 64, 1024$. The retained error metric is half-diamond distance of the fixed-launch preparation channel to $|d, 0\rangle\langle d, 0| \otimes \text{id}_B$. Because the dynamics factor as a latch preparation tensored with identity data, this distance equals the latch trace distance for arbitrary data references. It does not cover superposed eligibility sectors or source-history correlations.

Twenty-four exact algebraic checks verify spectra, stationarity, commutants, energy charges, encoded Pauli coefficients and spectator identities. A total of 262 binary64 checks verify positivity, normalization, cold no-jump relations, stationary convergence at sampled times, a blocked-ingress negative control and Bell-data retention. Finite tests do not certify the microscopic validity of the master equation.

At $r = 0$, the only populated live states are $m0, e0, d1$. Writing amplitudes as $(a, -ib, -c)$ gives the real no-jump generator

$$A = \begin{pmatrix} 0 & -g & 0 \\ g & 0 & -\lambda \\ 0 & \lambda & -\kappa/2 \end{pmatrix}. \quad (4)$$

TABLE I
COLD RETAINED-DOMAIN WINDOW BOUND AND INTERACTION BUDGET

κ	q_{CR} upper	$\ H_{CR}\ $	Unit-norm center time
2	$9.293 \cdot 10^{-5}$	1.5538	24.8604
8	$1.404 \cdot 10^{-15}$	2.5801	41.2814
32	$4.771 \cdot 10^{-11}$	4.8608	77.7729
128	$2.204 \cdot 10^{-10}$	9.5661	153.0571

Every downward jump parks the state in $d0$. Thus the exact fixed-launch retained error is $q_{CR} = a^2 + b^2 + c^2$; tracing the auxiliary gives $q_C = a^2 + b^2$. The distinction charges auxiliary excitation rather than silently discarding it.

The original window is $|t - 16| \leq 1/1024$. An 80-decimal-digit directed interval calculation encloses e^{At} over the entire window, using Taylor order 32 at scale 2^{14} , an analytic infinity-norm remainder, and interval squaring. Table I gives conservative upper bounds. The negative outcome at $\kappa = 2$ is retained; it is not replaced by a fitted success. The last three values pass 2^{-16} . The exceptional small value at $\kappa = 8$ is a conditional finite-model result, not an optimality claim or an independent performance prediction.

IV. REVERSE JUMPS: EXACT STATIONARY COST

For the defined local generator, a stationary density operator in order $(m0, m1, e0, e1, d0, d1)$ is

$$\rho_{ss} = \frac{\text{diag}(r, r^2, r, r^2, 1, r)}{(1+r)(1+2r)}. \quad (5)$$

Direct symbolic substitution verifies $[H_{CR}, \rho_{ss}] = 0$ and cancellation of the two dissipators. The remaining qutrit commutant of $S, S^\dagger$ and the m - e coupling is scalar. Numerical evolution converges to this state at the declared long deadline; no general all-time error floor is inferred from stationary values.

The exact stationary target errors are

$$\delta_{C,ss} = \frac{2r}{1+2r}, \quad (6)$$

$$\delta_{CR,ss} = \frac{3r+2r^2}{(1+r)(1+2r)}. \quad (7)$$

The stronger metric charges the auxiliary ground-state requirement. For a stationary retained target below 2^{-16} it is necessary that

$$2(65535)r^2 + 3(65535)r < 1. \quad (8)$$

For the clock-only target the corresponding condition is $r < 1/[2(65535)]$. Since $65535 = 3 \cdot 5 \cdot 17 \cdot 257$, these denominators are exact rate-budget arithmetic, not new geometric invariants. In particular $r = 2^{-16}$ fails both stationary targets, while $r = 2^{-20}$ passes them. Positive reverse rate also immediately destroys invariance of $|d, 0\rangle$: $J_\uparrow|d, 0\rangle \neq 0$.

Data preservation is a different statement. With identity-on-data coupling, the stored register and its reference remain unchanged even when the clock reactivates. Real devices must separately test deviations from that assumption; a simulated Bell control does not establish noise immunity.

V. CONSERVATIVE FINITE-WINDOW REVERSE-RATE BRIDGE

A derived bound extends the selected cold cell without a thermal trajectory fit. For $\Delta\mathcal{L} = \kappa r \mathcal{D}[I_C \otimes \sigma_R^\dagger]$, the diamond norm obeys $\|\Delta\mathcal{L}\|_\diamond \leq 2\kappa r$. Duhamel's formula and contraction of completely positive trace-preserving semigroups give a half-diamond perturbation debit at most $\kappa r t$.

For $\kappa = 8$, the interval cold bound is strictly below 2^{-48} . Choosing the conservative sufficient range $0 \leq r \leq 2^{-24}$ yields throughout the full window

$$\begin{aligned} \delta_{CR}(t) &\leq 2^{-48} + 8 \cdot 2^{-24}(16 + 1/1024) \\ &= \frac{2147614721}{281474976710656} < \frac{1}{65536}. \end{aligned} \quad (9)$$

This is a source-scoped sufficient model certificate, not an optimized thermal threshold. The range is selected from a derived bound after the cold evaluation, with that interpretation disclosed. It neither calibrates reservoir temperature nor closes physical promotion.

VI. ENERGY AND FULL-RESOURCE BOUNDARY

The full interaction spectrum is $0, 0, \pm g, \pm\sqrt{g^2 + \lambda^2}$. Its minimum centered norm is therefore $\sqrt{g^2 + \lambda^2}$; shifting energy cannot remove this width. Increasing κ at fixed Γ_{eff} increases $\lambda \propto \sqrt{\kappa}$. Equivalent unit-norm dynamics require scaling both Hamiltonian and dissipative rates by the same factor. Table I's time inflation concerns the interaction-picture budget only; dissipative-rate ceilings and laboratory free energies require separate accounting.

A candidate resonant energy assignment is $H_0 = \hbar\omega[(P_m + P_e) \otimes I + I \otimes N_R]$. The interaction commutes with H_0 , while bath lowering and raising change its energy by $\mp\hbar\omega$. Under this declared assignment, a cold capture exports one such quantum; this is not a universal dissipation lower bound. The local thermal generator is a model assumption, not a derivation of a global Gibbs bath for the dressed interacting Hamiltonian.

The next gate is an explicit admission/completion compiler under a matched full spectrum and rate budget. In parallel, audit a concrete bath spectral density and reverse-rate calibration. No autonomous completion detector, graphene capacity trigger, complete qubit two-local device, hardware validation or quantum advantage is promoted here. Physical promotion remains zero.

REFERENCES

- [1] F. Reiter and A. S. Sørensen, "Effective operator formalism for open quantum systems," *Phys. Rev. A*, vol.85, 032111, 2012. Author preprint: arXiv:1112.2806v2.